

Modern Software Development Methodologies

AGILE

Lesson Learning Objectives

At the end of the lecture, students should be able to:

- 1.Explain the concept of Agile software development and its advantages.
- 2.Identify the key values of the Agile Manifesto.
- 3.Recognize different Agile development methodologies.
- 4.Describe the Scrum framework, including roles, artifacts, and activities.
- 5.Write simple user stories using the standard user story format.

Content

- Traditional Vs Modern Methodologies
- Why Agile?
- Agile's Role in Different Domains
- Agile Development Methodologies
- SCRUM

Why Traditional Development Can Be Challenging

- Imagine a project that takes 12 months to complete.
- If customer requirements change after 6 months:
 - Rework is required
 - Cost increases
 - Delays occur
 - Customer may not get what they actually need

Example

- A customer requests an online shopping system.
- After 6 months, they realize they need mobile support and online payments.
- In a traditional approach, these changes can be costly and difficult to implement.
- **Agile was introduced to handle changing requirements more effectively.**

Traditional Vs Modern Methodologies

Traditional Way

- High Cost
- Changes are not acceptable
- Can detect errors only in the latter part of the SDLC
- Less or no iterations
- Lack of transparency

Modern Way

- Highest Value
- High Quality
- Shortest Time

Why Agile?

Challenges in Traditional Software Development

- Requirements may change during development.
- Customers may not see the system until the end of the project.
- Errors may be discovered late.
- Changes can be expensive to implement.
- Teams may build features that users no longer need.

How Agile Helps

- Delivers software in small increments.
- Encourages regular customer feedback.
- Adapts to changing requirements.
- Identifies issues early.
- Focuses on delivering high-value features first.

Agile helps teams deliver better software faster while responding to change.

Agile Development Methodologies

What is Agile?

- Agile Software Development is an umbrella term for a set of methods and practices based on the **Values** and **Principles** expressed in the Agile Manifesto
- The Agile Manifesto defines **4 key Values** and **12 Principles** that guide Agile software development.
- The 4 Values define the core philosophy of Agile, while the 12 Principles provide guidance on how these values are applied in practice.

4 Key Values

1. Individuals and Interactions

2. Working Software

3. Customer Collaboration

4. Responding to Change

4 Key Values Vs Traditional

In Agile	In Traditional Ways
1. Individuals and Interactions	Over Process and Tools
2. Working Software	Over Comprehensive Documentation
3. Customer Collaboration	Over Contract Negotiation
4. Responding to Change	Over Following Plan

Agile Methodologies and Practices

Agile Methodologies

Methods used to manage Agile projects.

- **Scrum** - Work is organized into Sprints.
- **Kanban** – Work is managed using a visual board.
- **Extreme Programming (XP)** – Focuses on engineering practices and code quality.
- **Lean Software Development** – Focuses on eliminating waste and maximizing value.

Agile Methodologies and Practices

Agile Practices

Techniques used while developing software.

- **Test-Driven Development (TDD)** - Write tests before writing code.
 - **Pair Programming** – Two developers work together on the same task.
 - **Behavior-Driven Development (BDD)** – Define system behavior using business-focused scenarios.
-
- Methodologies provide the overall structure for managing a project, while practices are specific techniques used by the development team during implementation.

SCRUM

In this lecture, we focus mainly on Scrum, which is the most widely used Agile methodology in industry.



What is SCRUM?

- Scrum is an Agile framework used to manage software development projects.
- It divides work into short iterations called **Sprints**.
- Teams deliver working software at the end of each Sprint.
- Feedback is collected regularly to improve the product.
- Scrum helps teams work collaboratively and adapt to change.

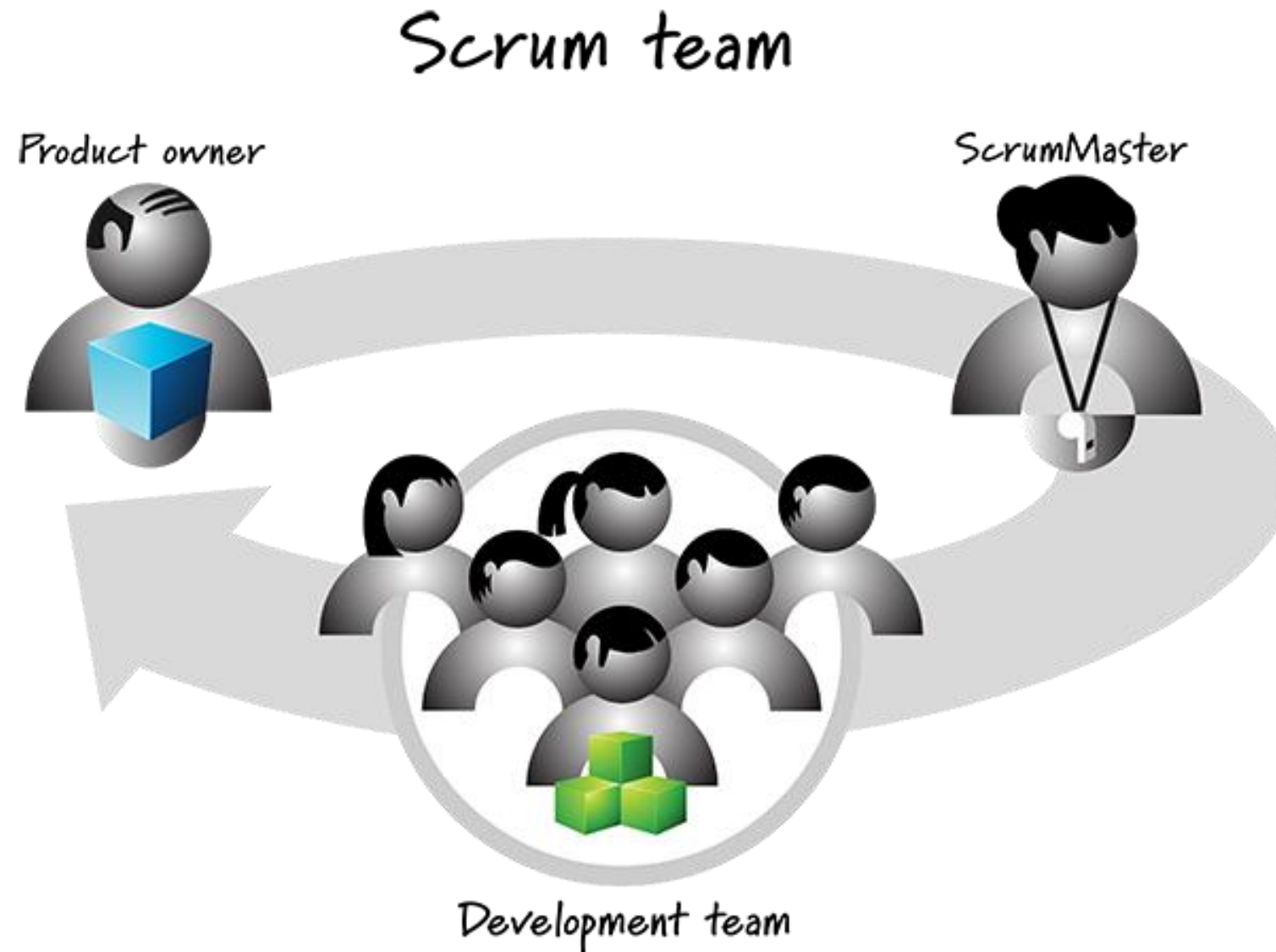
Scrum consists of Roles, Artifacts, and Activities.

SCRUM Roles

- A Scrum Team consists of three main roles:
 - **Product Owner** – Represents customers and prioritizes requirements.
 - **Scrum Master** – Ensures Scrum practices are followed and supports the team.
 - **Developers** – Build and deliver the product.

These roles work together to deliver working software at the end of each Sprint.

SCRUM Roles



SCRUM Roles

Product Owner

The Product Owner represents the customer and stakeholders.

Responsibilities

- Define and prioritize requirements.
- Maintain the Product Backlog.
- Communicate customer needs to the team.
- Decide what features should be developed.
- Accept or reject completed work.

Key Focus: Maximizing the value of the product.

SCRUM Roles

Scrum Master

The Scrum Master helps the team follow Scrum practices effectively.

Responsibilities

- Facilitate Scrum activities.
- Support the Scrum Team.
- Remove obstacles that affect progress.
- Promote collaboration and communication.
- Ensure Scrum principles are followed.

Key Focus: Helping the team work effectively.

Note: The Scrum Master is a facilitator, not a traditional project manager.

SCRUM Roles

Developers

Developers are responsible for building the product.

Responsibilities

- Design, develop, and test software.
- Complete Sprint Backlog items.
- Collaborate with other team members.
- Maintain product quality.
- Deliver a working product increment at the end of each Sprint.

Key Focus: Delivering working software.

SCRUM Artifacts

- Scrum uses artifacts to provide transparency and help teams manage their work.
- The three main Scrum artifacts are:
 - Product Backlog
 - Sprint Backlog
 - Burn Down Charts
- Artifacts help the Scrum Team understand what needs to be done and how much progress has been made.

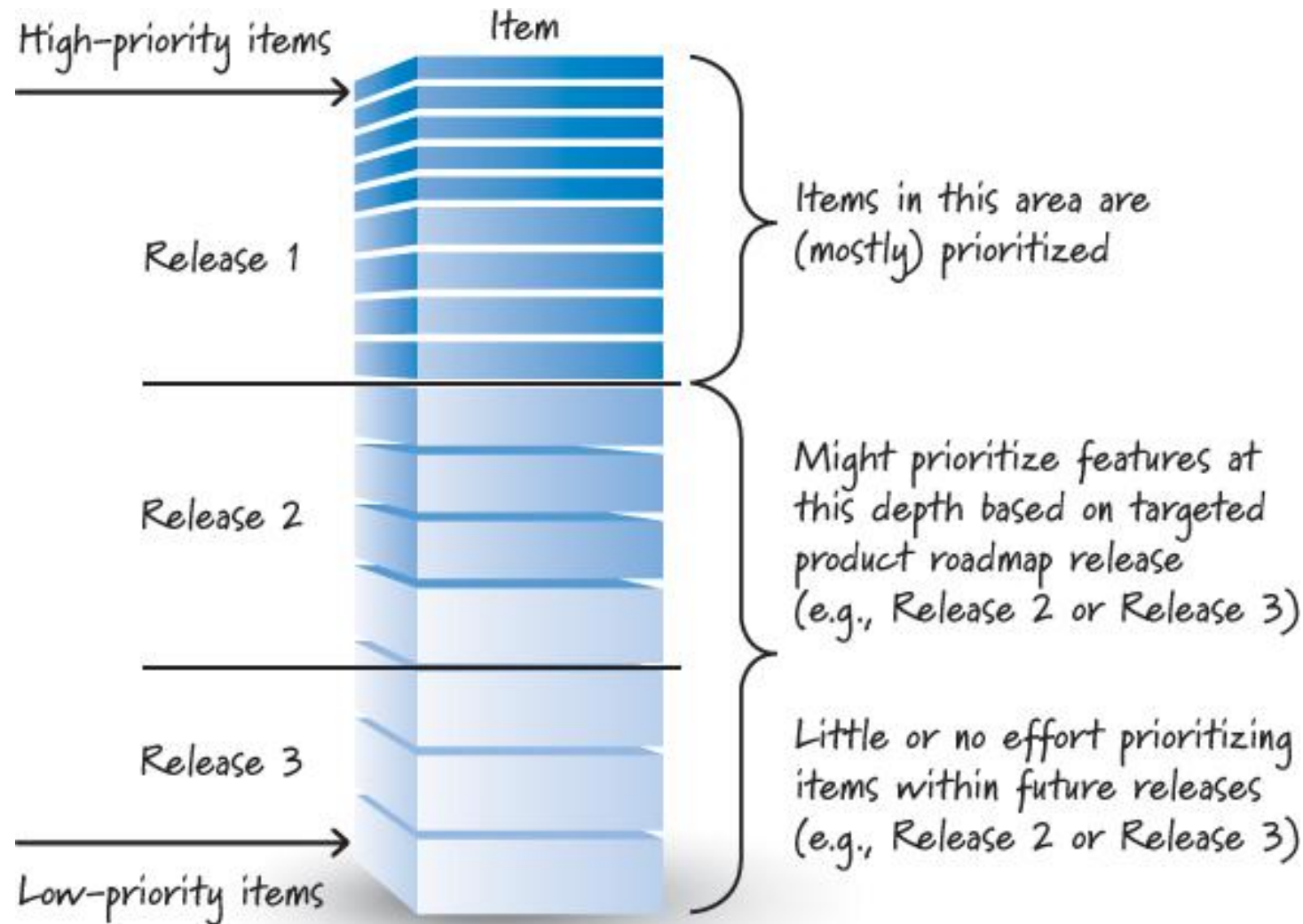
Product Backlog

- In Scrum, work is prioritized so that the highest-value items are completed first.
- The **Product Owner** is responsible for creating and maintaining the **Product Backlog**.
- The Product Backlog is a prioritized list of requirements, features, and improvements for the product.
- There are many ways to store the product backlog:
 - As a collection of index cards or post-its on the wall
 - In Excel
 - Write as user stories

Product Backlog

- Product Backlog items are placed in the correct sequence.
- High Value items appear at the top of the product backlog.
- Lower-valued items appear towards the bottom.
- The Product Backlog changes and evolves throughout the project.

Product Backlog

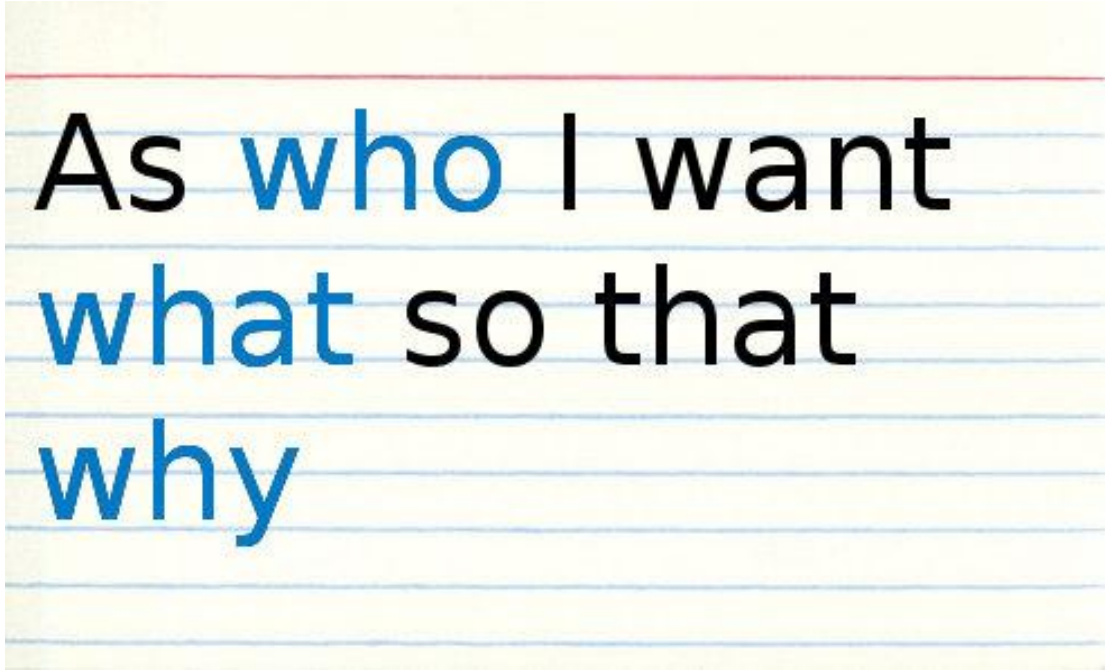


User Story

- This is a concise description of a **Functionality** that will be valuable to users of the system
- User stories are a convenient format for expressing the desired business value for many types of product backlog items
- User stories are crafted in a way that makes them understandable to both business and technical people

User Story Template

As a <user role> I want to <goal> So that <benefit>



As who I want
what so that
why

Writing User Stories

Example – Online Banking System

- **As a** Customer **I want to** view account summary online **so that** I do not have to wait till the month end to view the statement.
- **As an** Employee **I want to** add new customers online **so that** it saves my time.
- **As a** User **I want to** update profile details **so that** my details are up-to-date

Activity 01

Write 5 detailed User Stories for Library Management System

Example:

➤ **As a member I want to** view my borrowing history **so that** I can easily get the details of the books I referred.

What Makes a Good User Story?

- Clearly identify the **user role**.
- Describe the **goal** the user wants to achieve.
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- Explain the **benefit** ("So that...").
- Avoid technical implementation details.

Activity 02

Example 01:

Compare these User Stories.

A	B
As a customer, I want to view my account balance so that I can monitor my finances.	As a customer, I want a database table to store account balances.

Example 02: Compare these User Stories

A	B
As a member, I want to search books so that I can find books quickly.	As a member, I want an SQL query to search books.

Example 03: Compare these User Stories

A	B
As a librarian, I want to generate overdue book reports so that I can track late returns.	As a user, I want to generate reports.

Example 04: Compare these User Stories

A	B
As a customer, I want to save my favourite products.	As a customer, I want to save my favourite products so that I can find them easily later.

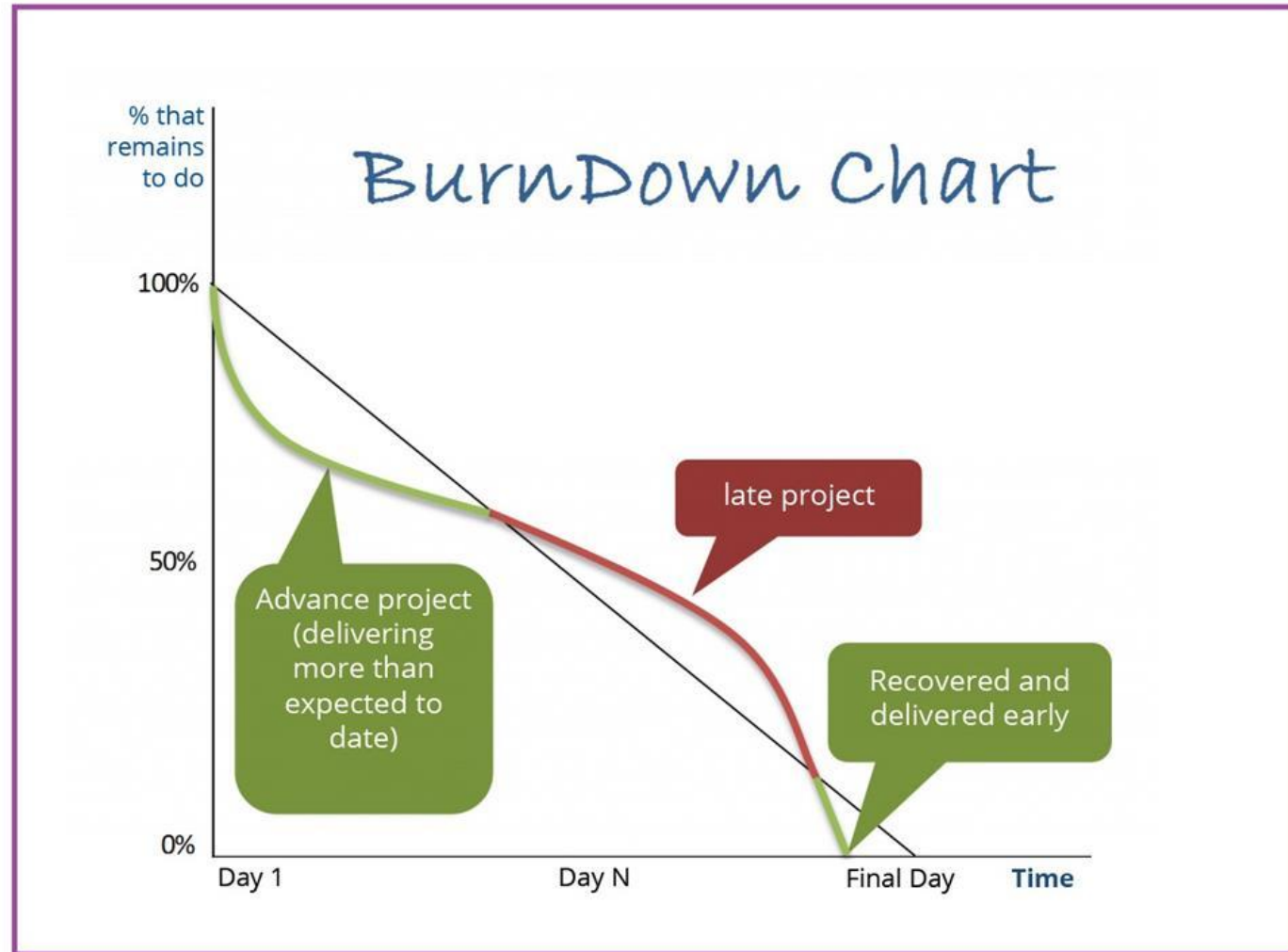
Sprint and Sprint Backlog

- In Scrum, work is performed in iterations or cycles of up to a calendar month called **sprints**
- Before starting a Sprint the Team should come up with a Sprint backlog
- The sprint backlog (release backlog) is a list of user stories identified by the Scrum team to be completed during the sprint
- This is a subset of Product backlog user stories defined only for a particular sprint

Sprint Burn Down Chart

- A burn down chart is a graphical representation of work left to do vs time.
- The outstanding work (or backlog) is often on the **vertical axis**, with time along the **horizontal**.
- That is, it is a run chart of outstanding work. It is useful for predicting when all of the work will be completed.

Sprint Burn Down Chart



Activity 03

- A software development team has **120 tasks** to complete within **5 days**.
- The number of tasks completed each day is shown below:

Days	01	02	03	04	05
Tasks	20	50	0	20	30

Tasks

- Calculate the **Ideal Burn Down Velocity** (tasks per day).
- Calculate the **Actual Burn Down Velocity** for each day.
- Identify the day with the **highest productivity** (most tasks completed).
- Identify the day with the **lowest productivity** (fewest tasks completed).
- Did the team complete all tasks on time?

SCRUM Activities

During each Sprint, the Scrum Team performs several activities to plan, monitor, review, and improve their work.

The main Scrum activities are:

- **Sprint Planning** – Decide what work will be completed during the Sprint.
- **Daily Scrum** - Monitor progress and discuss obstacles.
- **Sprint Review** - Demonstrate completed work and gather feedback.
- **Sprint Retrospective** - Reflect on the team's performance and identify improvements.

Sprint Planning

Sprint Planning is conducted at the beginning of each Sprint.

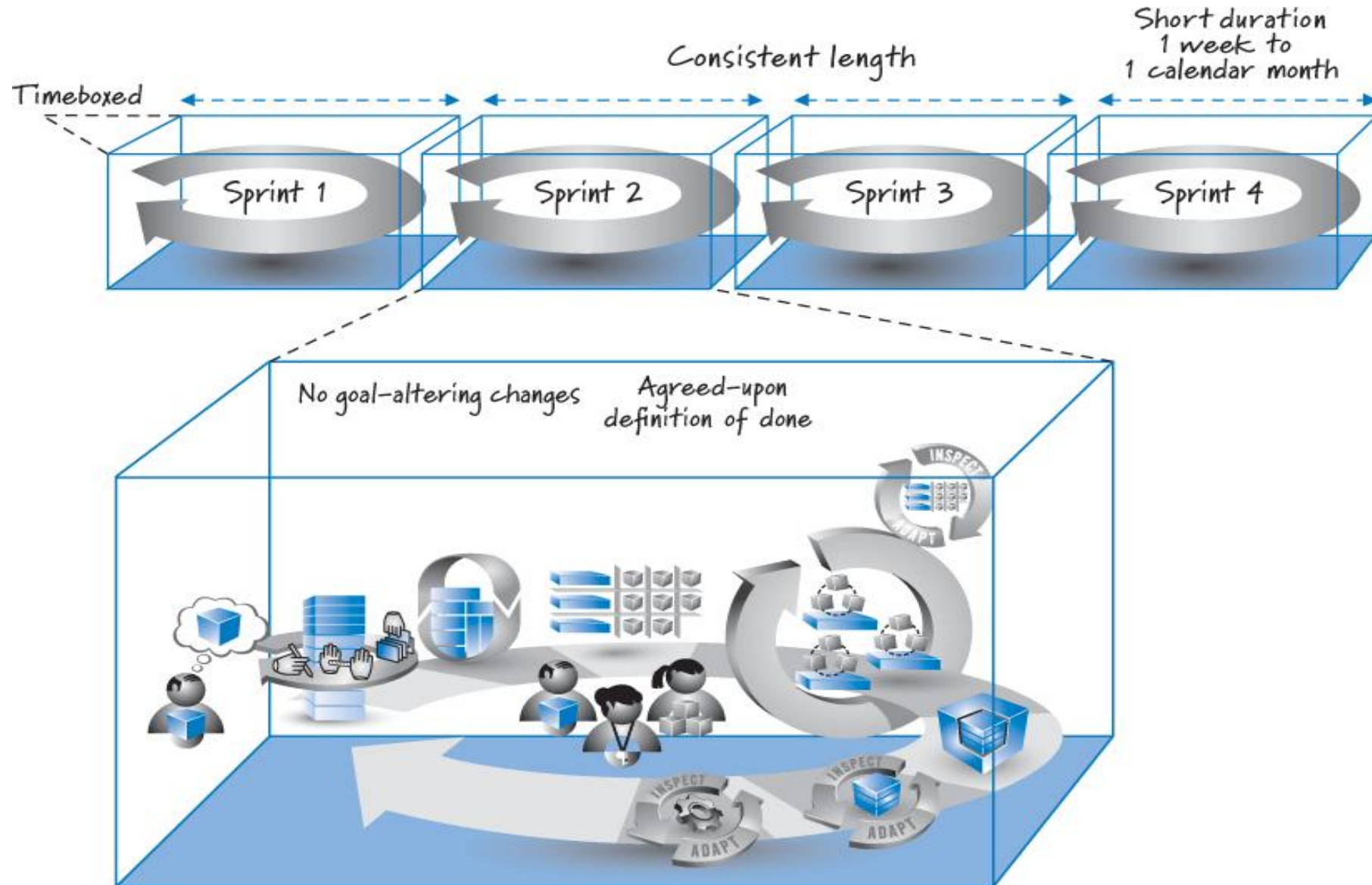
Purpose

- Select Product Backlog items for the Sprint.
- Define the Sprint Goal.
- Create the Sprint Backlog.

Outcome

A clear plan for the upcoming Sprint.

Sprint Planning

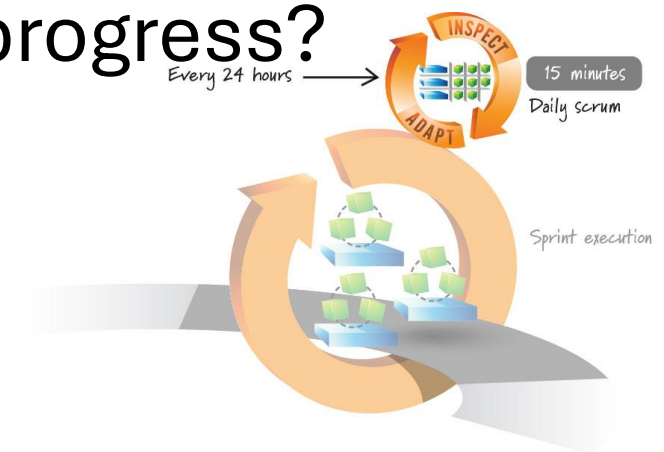


Daily Scrum

A short daily meeting (typically 15 minutes) held by the Scrum Team to monitor progress.

Three Key Questions

1. What did I complete since the last Daily Scrum?
2. What will I work on next?
3. Are there any obstacles blocking my progress?



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Sprint Review

Conducted at the end of the Sprint.

Purpose

- Demonstrate completed work.
- Gather stakeholder feedback.
- Review progress toward the Sprint Goal.
- Update the Product Backlog if needed.

Focus: The Product

Sprint Retrospective

Conducted after the Sprint Review.

Purpose

- Reflect on the Sprint.
- Identify what went well.
- Identify areas for improvement.
- Plan improvements for the next Sprint.

Focus: The Team and Process

Scrum Activities Summary

Activity	Main Purpose
Sprint Planning	Plan the Sprint
Daily Scrum	Monitor progress
Sprint Review	Review the Product
Sprint Retrospective	Improve the Process

Activity 04

What is the primary goal of the Sprint Review?

- 1) To plan the next Sprint
- 2) To inspect the Increment and adapt the Product Backlog if needed
- 3) To assign new tasks to team members
- 4) To conduct a retrospective on team performance

Activity 05

What is the purpose of the Sprint Retrospective?

- 1) To review the product increment and get feedback from the stakeholder
- 2) To plan the work for the next sprint and prioritize the product backlog
- 3) To inspect the team's performance and identify improvements for the next sprint
- 4) To coordinate the work across multiple teams and align on the product vision

Example: Sprint Planning and Backlog Management for a Hospital Appointment Management System

Scenario: Hospital Appointment Management System

Project Overview

- A private hospital wants to develop a web-based **Hospital Appointment Management System** that allows patients to book appointments with doctors online.
- Currently, patients must visit or call the hospital to make appointments, which often results in long waiting times and scheduling conflicts.
- The objective is to provide an online system that simplifies appointment booking, improves communication, and reduces administrative work.

Product Backlog

ID	User Story	Priority
PBI-01	As a patient, I want to register an account so that I can use the appointment system.	High
PBI-02	As a patient, I want to search for doctors by specialization so that I can find the correct doctor.	High
PBI-03	As a patient, I want to book an appointment so that I can reserve a consultation time.	High
PBI-04	As a doctor, I want to view my daily appointments so that I can prepare for consultations.	Medium
PBI-05	As a patient, I want to receive appointment reminders so that I do not miss my appointment.	Low

Sprint Planning

Sprint Planning

1st Sprint (Weeks 1–2)

- The development team may select some of the highest-priority user stories.

Sprint ID	User Story	Priority
SP1-01	As a patient, I want to register an account so that I can use the appointment system.	High
SP1-02	As a patient, I want to search for doctors by specialization so that I can find the correct doctor.	High

Why were these User Stories Selected for Sprint 1?

Sprint 1 Goal

Deliver a Minimum Viable Product (MVP) that allows patients to start using the system.

Selection Criteria

- ✓ Highest Business Value
 - These features provide the greatest value to users and the hospital.
- ✓ Core System Functionality
 - Registration, doctor search, and appointment booking are essential for the system to operate.
- ✓ Foundation for Future Features
 - Other features depend on these core functions being completed first.
- ✓ Deliver a Working Product Early
 - At the end of Sprint 1, patients can successfully use the basic appointment system.

Why were Other Features Deferred?

- Appointment reminders and doctor dashboards improve the system but are not essential for the first working version.
- They can be implemented in later sprints after the core functionality is completed.

Key Scrum Principle

Always develop the highest-value features first and deliver working software in small, incremental releases.

Sprint Backlog

Task Breakdown for User Story 1 (Patient Registration)

Task ID	Task	Est. Hours
T-01.1	Design patient registration page	5
T-01.2	Develop registration backend API	6
T-01.3	Validate patient information	3
T-01.4	Save patient details to the database	3
T-01.5	Test registration functionality	3

Sprint Backlog

Task Breakdown for User Story 2 (Doctor Search)

Task ID	Task	Est. Hours
T-02.1	Design doctor search page	4
T-02.2	Develop doctor search API	6
T-02.3	Implement search filters	4
T-02.4	Retrieve doctor availability	3
T-02.5	Test search functionality	3

Sprint Planning

Sprint Planning

2nd Sprint (Weeks 3–4)

- The development team may select some of the highest-priority user stories.

Sprint ID	User Story	Priority
SP1-01	As a patient, I want to book an appointment so that I can reserve a consultation time.	High
SP1-02	As a doctor, I want to view my daily appointments so that I can prepare for consultations.	Medium

Sprint Backlog

Task Breakdown for User Story 3 (Appointment Booking)

Task ID	Task	Est. Hours
T-03.1	Design appointment booking page	5
T-03.2	Develop appointment booking API	6
T-03.3	Validate doctor availability	4
T-03.4	Save appointment details	3
T-03.5	Test appointment booking functionality	3

Sprint Backlog

Task Breakdown for User Story 4 (Doctor Appointment View)

Task ID	Task	Est. Hours
T-04.1	Design doctor's appointment dashboard	4
T-04.2	Develop appointment retrieval API	5
T-04.3	Display daily appointment schedule	4
T-04.4	Add appointment status indicators	3
T-04.5	Test appointment dashboard	3

Remaining Backlog: Lower-priority item left for a future sprint

Discussion Activity

Scenario

The hospital management has decided that appointment reminders must be available before the system is launched because many patients miss their appointments.

Discussion Questions

1. Which Product Backlog Item should have its priority changed?
2. What should its new priority be and Why?

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Thank You